

SIMATS ENGINEERING

ASSESSMENT TOOL 2: RESEARCH REVIEW & ANALYSIS

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ASSESSMENT TOPIC

Decision Tree Learning: A Research Review of ID3, C4.5 and CART with Prolog-Based Analysis

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STUDENT DECLARATION

I, S. Roshni (192525185), certify that the submitted work is my original work and that I have followed the academic integrity guidelines specified for this assessment.

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TITLE

Decision Tree Learning: Comparative Research Review of ID3, C4.5 and CART Using Prolog-Based Decision Analysis

ABSTRACT

Decision Trees are important supervised learning techniques used for classification and prediction. This review analyzes three classical decision-tree algorithms: ID3, C4.5 and CART. The algorithms are compared based on attribute-selection methods, pruning strategies, data handling, interpretability, computational requirements and classification performance.

ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART commonly uses the Gini Index for classification. Research shows that C4.5 improves several limitations of ID3 by supporting continuous attributes and pruning, while CART constructs binary trees and uses complexity-based pruning.

The review identifies a research gap in the dependence of traditional decision trees on greedy, single-attribute splitting criteria. A possible improvement is to combine impurity measures with feature-weighting, statistical stopping criteria and explainable rule generation. A small Prolog-based implementation is proposed to demonstrate how learned decision rules can be represented and queried.

KEYWORDS

Decision Tree, ID3, C4.5, CART, Gini Index, Information Gain

1. INTRODUCTION

Decision Tree Learning is a supervised machine-learning approach that represents classification decisions in the form of a tree. Internal nodes represent tests on attributes, branches represent outcomes of tests, and leaf nodes represent final classes.

Decision trees are popular because their predictions can be converted into understandable decision rules. They are therefore useful in applications such as loan approval, medical diagnosis, fraud detection, customer classification and recommendation systems.

Three important decision-tree algorithms are ID3, C4.5 and CART. ID3 introduced an information-gain-based approach, C4.5 extended ID3 with gain ratio and additional data-handling capabilities, while CART introduced binary tree construction using criteria such as the Gini Index.

This review focuses on the differences among these three algorithms and considers how their decision rules can be represented using Prolog.

2. RESEARCH PROBLEM AND OBJECTIVES

Research Problem

Although ID3, C4.5 and CART all construct decision trees, they differ in their splitting criteria, tree structure, pruning methods and ability to handle different types of data. Selecting the appropriate algorithm depends on the characteristics of the dataset and the requirements of the application.

Objectives

- To study ID3, C4.5 and CART.
- To compare their attribute-selection methods.
- To analyze their pruning strategies.
- To compare interpretability and computational requirements.
- To examine their classification performance.
- To identify limitations of traditional decision-tree algorithms.
- To identify a research gap.
- To propose an improved decision-tree research direction.
- To demonstrate decision rules using Prolog.

3. LITERATURE REVIEW

3.1 ID3

ID3 is one of the early decision-tree algorithms. It uses Entropy and Information Gain to select the attribute that provides the highest reduction in uncertainty.

The basic entropy formula is:

$$\text{Entropy}(S) = -\sum p_i \log_2(p_i)$$

Information Gain is calculated as:

$$\text{Gain}(S,A) = \text{Entropy}(S) - \sum (|S_v| / |S|) \text{Entropy}(S_v)$$

ID3 is relatively simple and produces interpretable trees. However, research indicates that the original ID3 does not provide pruning and has limitations with numeric attributes and missing values.

Another important limitation is that Information Gain can favor attributes having many distinct values. This bias has been discussed in research on decision-tree splitting criteria.

3.2 C4.5

C4.5 was developed as an extension of ID3 by Ross Quinlan. It uses Gain Ratio instead of directly using Information Gain.

The basic form is:

$$\text{Gain Ratio} = \text{Information Gain} / \text{Split Information}$$

C4.5 can handle continuous attributes and incorporates pruning after tree construction. Research descriptions of C4.5 show that error-based pruning is applied after the tree-growing stage.

C4.5 therefore addresses several limitations of ID3 and generally produces more controlled trees.

Quinlan's original C4.5 work describes the system as producing decision trees or if-then rules from cases containing nominal and numeric properties, with emphasis on understandable and accurate models.

3.3 CART

CART stands for Classification and Regression Trees.

For classification, CART commonly uses the Gini Index:

$$\text{Gini} = 1 - \sum p_i^2$$

Unlike the multiway structure commonly associated with ID3 and C4.5, CART constructs binary decision trees. CART also uses pruning based on tree complexity.

CART is widely used because binary splitting, impurity reduction and pruning provide a practical framework for classification and regression.

4. COMPARATIVE ANALYSIS

Parameter	ID3	C4.5	CART
Main criterion	Information Gain	Gain Ratio	Gini Index
Tree structure	Multiway	Multiway / binary for numeric splits	Binary
Numeric data	Limited	Supported	Supported
Missing values	Limited	Better handling	Supported in implementations
Pruning	No original pruning	Error-based pruning	Complexity pruning
Interpretability	High	High	High
Main objective	Classification	Classification	Classification and Regression
Splitting approach	Greedy	Greedy	Greedy binary
Main limitation	Multi-value bias	More computation than basic ID3	Binary tree can become larger

The core differences in splitting criteria and tree structures are consistently described in decision-tree literature.

5. ATTRIBUTE-SELECTION METHODS

ID3 – Information Gain

Information Gain measures the reduction in entropy after splitting the dataset.

Advantage: Simple and effective.

Limitation: Can favour attributes with many possible values.

C4.5 – Gain Ratio

Gain Ratio modifies Information Gain by considering the intrinsic information generated by a split.

Advantage: Reduces the multi-valued attribute bias of Information Gain.

Limitation: Gain Ratio can still have statistical biases under some circumstances. Research has identified biases affecting gain-ratio and related splitting measures.

CART – Gini Index

Gini Index measures node impurity.

Advantage: Computationally convenient for classification and commonly used in CART implementations.

Limitation: Like other greedy splitting criteria, it evaluates candidate splits mainly through local information rather than considering larger combinations of features.

6. PRUNING STRATEGIES

Pruning is important because a very large decision tree may memorize training data and suffer from overfitting.

ID3

The original ID3 does not perform a formal pruning process.

C4.5

C4.5 uses post-tree construction pruning. Subtrees can be replaced by leaf nodes when pruning does not reduce predictive performance significantly.

CART

CART uses a complexity-based pruning strategy to control tree size.

7. INTERPRETABILITY

Interpretability is one of the strongest advantages of decision trees.

For example:

```
IF Credit_Score >= 700 AND Repayment_History = Good AND Income >= 40000 THEN Loan = Approved
```

Such rules can be easily understood by a human.

C4.5 is particularly relevant because its models can also be represented as decision trees or if-then rules.

For Prolog, these rules can be represented directly as logical predicates.

8. COMPUTATIONAL COMPLEXITY

Decision-tree algorithms generally use a greedy process:

Select best attribute	↓	Split data	↓	Repeat for child nodes
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The computational cost depends on the number of instances, attributes, possible split points and depth of the tree.

ID3 has a relatively simple calculation but may require repeated entropy calculations. C4.5 adds gain-ratio calculations and support for continuous attributes. CART searches binary splits and therefore has additional split-selection operations.

Research comparing splitting criteria shows that the choice of splitting measure can affect classification accuracy, tree depth, leaf count and construction time.

9. CLASSIFICATION PERFORMANCE

There is no single decision-tree algorithm that is always best for every dataset.

Performance depends on:

- Dataset size
- Number of attributes
- Noise
- Missing values
- Class imbalance
- Attribute cardinality
- Pruning strategy
- Hyperparameter settings

A comparative study examining ID3, C4.5 and CART highlights differences in their working processes and performance across datasets.

Therefore, algorithm selection should be based on experimental evaluation rather than assuming that one method will always produce the highest accuracy.

10. CRITICAL ANALYSIS

Strengths of ID3

- Simple algorithm.
- Easy to understand.
- Effective for basic categorical datasets.
- Provides clear decision rules.

Weaknesses of ID3

- No original pruning.
- Information Gain can favor high-cardinality attributes.
- Limited support for continuous data.
- Sensitive to noisy training data.

Strengths of C4.5

- Uses Gain Ratio.
- Supports continuous attributes.
- Provides pruning.
- Produces interpretable rules.

Weaknesses of C4.5

- More computationally complex than basic ID3.
- Gain Ratio does not completely remove splitting bias.

- Large trees may still require careful pruning.

Strengths of CART

- Uses Gini Index for classification.
- Produces binary trees.
- Supports classification and regression.
- Uses complexity-based pruning.

Weaknesses of CART

- Binary splitting can create deeper trees.
- Performance depends strongly on split and pruning settings.
- Greedy splitting may miss useful feature interactions.

11. RESEARCH GAP

The review identifies the following research gaps:

- Traditional decision trees mainly use greedy single-attribute splitting.
- Information Gain and Gain Ratio can show bias under particular attribute distributions.
- Gini-based splitting also relies on local impurity reduction.
- Decision trees can become unstable when training data changes.
- Large trees can reduce interpretability.
- Static decision trees are less suitable for continuously changing data.

Research has proposed alternatives that consider higher-order relationships rather than only individual feature-target correlations. One such research direction develops splitting criteria based on correlations involving small subsets of attributes.

A further research opportunity is therefore to combine multi-feature relationships, pruning and explainable Prolog rules.

12. PROPOSED RESEARCH DIRECTION

Proposed Model: Hybrid Explainable Decision Tree with Prolog

The proposed system combines:

Dataset	↓ Feature Selection	↓ Information Gain / Gini Analysis	↓ Multi-
feature Evaluation	↓ Decision Tree	↓ Pruning	↓ Prolog Rule Generation
Explainable Prediction			↓

Proposed Improvements

- Combine Gini and Information Gain.
- Use feature weighting before splitting.
- Apply stronger pruning.
- Detect class imbalance.

- Consider combinations of important attributes.
- Convert the final tree into Prolog rules.
- Provide an explanation for every prediction.

This can improve interpretability while reducing unnecessary tree complexity.

13. PROLOG-BASED ANALYSIS

Prolog is suitable for representing the final decision rules because decision trees can naturally be expressed as logical conditions.

Prolog Facts

```
customer(1, 60000, 780, employed, 250000, good). customer(2, 25000, 580, unemployed, 500000, poor). customer(3, 50000, 720, employed, 200000, good).
```

Decision Rules

```
loan_decision(Income, Credit, employed, Amount, good, approved) :-
Credit >= 700, Income >= 40000, Amount =< Income * 6. loan_decision(_,
Credit, _, _, rejected) :- Credit < 650. loan_decision(_, _, _, poor,
rejected).
```

Prediction Query

```
?- loan_decision(        60000,        780,        employed,        250000,        good,
Result    ).
```

Output

```
Result = approved.
```

Thus, Prolog provides a transparent logical representation of the final decision rules.

14. RESEARCH METHODOLOGY / ANALYSIS FRAMEWORK

The proposed research follows these stages:

1. Literature Collection	↓ 2. Algorithm Identification	↓ 3. ID3 /
C4.5 / CART Comparison	↓ 4. Attribute-Selection Analysis	↓ 5.
Pruning Analysis	↓ 6. Performance Analysis	↓ 7. Research Gap
Identification	↓ 8. Proposed Hybrid Approach	↓ 9. Prolog Rule
Representation	↓ 10. Experimental Validation	

The review compares the algorithms using:

- Attribute-selection criterion
- Tree structure
- Pruning
- Interpretability
- Computational cost
- Data handling

- Classification performance
- Generalization

15. EXPERIMENTAL / EVIDENCE-BASED VALIDATION

A future experimental study can use standard classification datasets and compare:

- ID3
- C4.5
- CART
- Proposed Hybrid Model

Evaluation Metrics

Metric	Purpose
Accuracy	Overall correct predictions
Precision	Correct positive predictions
Recall	Correctly identified positive cases
F1-score	Balance between precision and recall
Tree Depth	Model complexity
Number of Nodes	Tree size
Training Time	Computational cost

Research comparing multiple splitting criteria has used metrics such as classification accuracy, tree depth, leaf-node count and tree-construction time.

16. DISCUSSION

The review shows that ID3, C4.5 and CART share the basic divide-and-conquer decision-tree principle but differ significantly in their splitting criteria and pruning strategies.

ID3 is the simplest but has important limitations. C4.5 improves ID3 by using Gain Ratio and supporting continuous attributes and pruning. CART provides binary splitting and uses Gini-based impurity reduction with complexity-based pruning.

For applications requiring transparency, all three approaches are useful. However, traditional greedy splitting can limit performance when useful information depends on interactions between multiple attributes.

A hybrid approach combining improved feature selection, pruning and Prolog-based rule representation could provide both predictive performance and explainability.

17. CONCLUSION

This research review analyzed ID3, C4.5 and CART, three important decision-tree learning algorithms.

ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART uses the Gini Index for classification. C4.5 and CART provide stronger mechanisms for handling practical datasets than the original ID3, particularly through support for continuous data and pruning.

The major research gap is the limitation of traditional greedy splitting approaches when important relationships involve multiple attributes. A possible future direction is a hybrid explainable decision-tree system that combines improved feature selection, pruning and multi-feature analysis.

Prolog can then be used to represent the resulting decision tree as understandable logical rules, making the system suitable for explainable AI applications.

18. REFERENCES

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BACK PAGE – RUBRICS EVALUATION

STUDENT EVALUATION SHEET

S.No.	Evaluation Criterion	Weightage	Marks Obtained
1	Problem Analysis and Domain Understanding	10%	
2	Literature Search and Source Selection	10%	
3	Literature Review and Synthesis	15%	
4	Comparative Analysis of AI Techniques	10%	
5	Critical Analysis of Research Findings	10%	
6	Research Gap Identification	10%	
7	Proposed Research Direction	10%	
8	Research Methodology / Analysis Framework	5%	
9	Experimental / Evidence-Based Validation	10%	
10	Result Analysis and Interpretation	5%	
11	Research Paper Quality and Referencing	10%	
12	Technical Presentation and Research Defense	5%	
	TOTAL	100%	____ / 100

FACULTY EVALUATION

Total Marks Awarded: _____ / 100

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